

MENTORA ACADEMY

DEPARTMENT OF ADVANCED CHEMISTRY · CAMBRIDGE INTERNATIONAL A LEVEL (9701)

Inorganic Chemistry Practical: Paper 5 Master Suite

10 Most Frequently Tested Planning, Analysis & Evaluation Questions (Past 4 Years Analysis: 2021–2024)

CANDIDATE NAME:	Urwah	CENTRE NUMBER:	PK082	CANDIDATE NO:	0142
COMPONENT:	Paper 5 — Planning, Analysis and Evaluation (Past 10 Years Archive)	DURATION:	2 Hours 30 Minutes	MAX MARKS:	100 Marks

READ THESE INSTRUCTIONS FIRST

- Write your name, Centre number, and candidate number in the boxes above.
- Answer all questions in the spaces provided on the question paper.
- In Question 1 (Planning): define independent, dependent, and controlled variables; describe step-by-step procedures with apparatus diagrams; calculate required quantities; state hazard risk assessments.
- In Question 2 (Analysis, Conclusions & Evaluation): process raw experimental data into derived columns; quote values to consistent significant figures; plot graphs with appropriate linear scales; determine gradients and intercepts; identify anomalous points; evaluate uncertainties.
- You may use an approved scientific calculator. The Periodic Table is provided as standard reference.

EXAMINATION BREAKDOWN & SYLLABUS SKILLS TESTED

Question	Investigation Title & Focus	Syllabus Ref	Type of Task	Marks
Question 1	Planning a Colorimetric Determination of Complex Formula via Job's Method	9701/52/M/J/24/Q1	PLANNING	10
Question 2	Analysis and Evaluation of the Stoichiometry of Malachite Thermal Decomposition	9701/51/O/N/23/Q1	ANALYSIS & EVALUATION	10
Question 3	Planning a Redox Titrimetric Analysis of Chromium Content in Ferrochrome Alloy	9701/53/M/J/23/Q2	PLANNING	10
Question 4	Planning the Synthesis and Recrystallisation of Hexaamminecobalt(III) Chloride	9701/52/O/N/22/Q1	PLANNING	10
Question 5	Analysis and Evaluation of the Stability Constant (Kstab) of Copper(II) Ammine Complex	9701/51/M/J/22/Q2	ANALYSIS & EVALUATION	10
Question 6	Planning Gravimetric Determination of Sulfate Content in Agricultural Fertilizer	9701/52/M/J/21/Q1	PLANNING	10
Question 7	Quantitative Analysis of Decomposition Trends in Group 2 Nitrates	9701/53/O/N/21/Q2	ANALYSIS & EVALUATION	10
Question 8	Planning a Potentiometric Redox Titration of Iron(II) with Permanganate	9701/51/M/J/20/Q1	PLANNING	10
Question 9	Analysis of Systematic Errors in the Iodometric Determination of Copper in Brass	9701/52/O/N/24/Q2	ANALYSIS & EVALUATION	10
Question 10	Planning the Synthesis and Anti-Solvent Precipitation of $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4 \cdot \text{H}_2\text{O}$	9701/53/M/J/24/Q1	PLANNING	10

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QUESTION 1

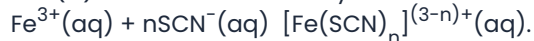
PLANNING

[Total: 10 Marks]

Planning a Colorimetric Determination of Complex Formula via Job's Method

Cambridge Syllabus Reference: 9701/52/M/J/24/Q1

Iron(III) ions react with thiocyanate ions in acidic solution to form an intense blood-red complex:



In Job's method of continuous variation, the total molar concentration ($[\text{Fe}^{3+}] + [\text{SCN}^{-}]$) is kept constant while the mole fraction of thiocyanate, $x_{\text{SCN}^{-}}$, is varied from 0.0 to 1.0.

Plan an experiment to determine the value of n using a colorimeter.

- (a) Specify the concentrations and volumes of equimolar solutions of $\text{Fe}(\text{NO}_3)_3$ and KSCN ($0.0020 \text{ mol dm}^{-3}$ in $0.1 \text{ mol dm}^{-3} \text{ HNO}_3$) to prepare seven mixtures of total volume 25.0 cm^3 . [3]

Mixture	Vol $\text{Fe}(\text{NO}_3)_3$ / cm^3	Vol KSCN / cm^3	$x(\text{SCN})$
1	22.5	2.5	0.10
2	17.5	7.5	0.30
3	12.5	12.5	0.50
4	10.0	15.0	0.60
5	7.5	17.5	0.70
6	5.0	20.0	0.80
7	2.5	22.5	0.90

- (b) State the appropriate colorimeter filter to choose and how to zero the instrument. [2]

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- (c) Describe how the value of n is determined from a plot of absorbance against $x_{\text{SCN}^{-}}$. [3]

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- (d) If the maximum absorbance occurs at $x_{\text{SCN}^{-}} = 0.50$, deduce the formula of the complex. [2]

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QUESTION 2

ANALYSIS & EVALUATION

[Total: 10 Marks]

Analysis and Evaluation of the Stoichiometry of Malachite Thermal Decomposition

Cambridge Syllabus Reference: 9701/51/O/N/23/Q1

Basic copper(II) carbonate, malachite, has the empirical formula $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$.

Upon strong heating in a porcelain crucible, it undergoes thermal decomposition according to:



Molar masses: $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2 = 221.0 \text{ g mol}^{-1}$; $\text{CuO} = 79.5 \text{ g mol}^{-1}$; $\text{CO}_2 = 44.0 \text{ g mol}^{-1}$; $\text{H}_2\text{O} = 18.0 \text{ g mol}^{-1}$.

- (a) Calculate the theoretical percentage mass loss expected when malachite decomposes completely. [2]
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-
- (b) A student heated 4.420 g of malachite and recorded a final constant mass of residue of 3.280 g. Calculate the experimental percentage mass loss. [2]
-
-
- (c) Suggest two distinct reasons why the experimental percentage mass loss is lower than theoretical. [2]
-
-
-
- (d) Explain the role of the crucible lid and why it must be slightly ajar during heating but closed during cooling. [2]
-
-
- (e) Suggest a chemical test to confirm that the black residue is pure CuO and contains no unreacted malachite. [2]
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QUESTION 3

PLANNING

[Total: 10 Marks]

Planning a Redox Titrimetric Analysis of Chromium Content in Ferrochrome Alloy

Cambridge Syllabus Reference: 9701/53/M/J/23/Q2

Ferrochrome is an alloy of iron and chromium. Plan an investigation to determine the percentage of chromium by mass in a sample of ferrochrome.

The alloy is dissolved in hot concentrated sulfuric acid to yield Fe^{2+} and Cr^{3+} .

Cr^{3+} is oxidised to $\text{Cr}_2\text{O}_7^{2-}$ using ammonium peroxodisulfate, $(\text{NH}_4)_2\text{S}_2\text{O}_8$, in the presence of a silver nitrate catalyst. The dichromate(VI) formed is titrated with standardized iron(II) ammonium sulfate.

- (a) Explain why the mixture must be boiled vigorously after adding $(\text{NH}_4)_2\text{S}_2\text{O}_8$ before titration begins. [2]
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-
- (b) Write the balanced ionic equation for the reaction between $\text{Cr}_2\text{O}_7^{2-}$ and Fe^{2+} in acidic solution. [2]
-
-
- (c) State a suitable indicator for this titration and state the colour change at the end-point. [2]
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-
- (d) Outline the mathematical steps to calculate the percentage by mass of Cr in the alloy from the mean titre of Fe^{2+} . [3]
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-
- (e) State the hazard associated with dichromate(VI) solutions and the corresponding safety measure. [1]
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QUESTION 4

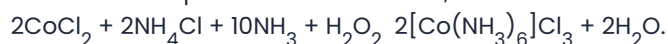
PLANNING

[Total: 10 Marks]

Planning the Synthesis and Recrystallisation of Hexaamminecobalt(III) Chloride

Cambridge Syllabus Reference: 9701/52/O/N/22/Q1

Hexaamminecobalt(III) chloride, $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ (orange–yellow crystals), is prepared by the oxidation of cobalt(II) chloride in the presence of ammonia, ammonium chloride, and activated charcoal catalyst:



The crude precipitate contains both $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ and charcoal.

- (a) Explain the role of activated charcoal in this preparation. [2]
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- (b) Describe how the crude product containing charcoal is separated from the reaction mixture. [2]
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-
- (c) Describe how pure $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ is separated from the charcoal catalyst during recrystallisation. [3]
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- (d) Explain why hot dilute HCl is used instead of pure water for dissolving the complex. [2]
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- (e) State how the crystals should be dried to obtain constant mass without decomposition. [1]
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QUESTION 5

ANALYSIS & EVALUATION

[Total: 10 Marks]

Analysis and Evaluation of the Stability Constant (K_{stab}) of Copper(II) Ammine Complex

Cambridge Syllabus Reference: 9701/51/M/J/22/Q2

The stepwise substitution of water by ammonia in aqueous copper(II) forms $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$:
 $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}(\text{aq}) + 4\text{NH}_3(\text{aq}) \rightleftharpoons [\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}(\text{aq}) + 4\text{H}_2\text{O}(\text{l})$.
The overall stability constant expression is: $K_{\text{stab}} = \frac{[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}}{[\text{Cu}^{2+}][\text{NH}_3]^4}$.

- (a) Explain why $[\text{H}_2\text{O}]$ is omitted from the K_{stab} expression. [2]

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- (b) In an equilibrium mixture, $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+} = 0.0480 \text{ mol dm}^{-3}$, $[\text{Cu}^{2+}] = 2.40 \times 10^{-14} \text{ mol dm}^{-3}$, and $[\text{NH}_3] = 0.250 \text{ mol dm}^{-3}$. Calculate K_{stab} and state its units. [3]

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- (c) Calculate the standard Gibbs free energy change, G° , for this complex formation at 298 K ($G^\circ = -RT \ln K_{\text{stab}}$). [2]

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- (d) Explain in terms of the chelate effect why the complex with 1,2-diaminoethane (en), $[\text{Cu}(\text{en})_3]^{2+}$, has a substantially higher K_{stab} than the tetraammine complex. [3]

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QUESTION 6

PLANNING

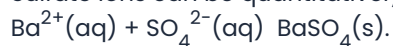
[Total: 10 Marks]

Planning Gravimetric Determination of Sulfate Content in Agricultural Fertilizer

Cambridge Syllabus Reference: 9701/52/M/J/21/Q1

Agricultural fertilizer contains ammonium sulfate, $(\text{NH}_4)_2\text{SO}_4$.

Sulfate ions can be quantitatively precipitated as insoluble barium sulfate:



Plan an experiment to determine the percentage by mass of sulfate in a fertilizer sample using gravimetric analysis.

- (a) Explain why the fertilizer solution must be acidified with dilute HCl before adding barium chloride. [2]
-
-
- (b) Describe the 'digestion' process after precipitation and explain why it is essential for accurate filtration. [2]
-
-
- (c) Specify the grade of filter paper required and describe how the precipitate is transferred without loss. [2]
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-
-
- (d) Describe how the precipitate is heated to constant mass in a crucible. [2]
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- (e) State how the student should test the filtrate to confirm that all sulfate has been precipitated. [2]
-
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QUESTION 7

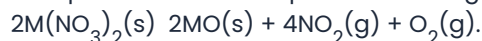
ANALYSIS & EVALUATION

[Total: 10 Marks]

Quantitative Analysis of Decomposition Trends in Group 2 Nitrates

Cambridge Syllabus Reference: 9701/53/O/N/21/Q2

Group 2 nitrates decompose on heating according to:



A student investigated the decomposition temperature of anhydrous nitrates of Mg, Ca, Sr, and Ba.

The table lists the ionic radius of M^{2+} and the minimum decomposition temperature T_{decomp} .

- (a) Complete the table by calculating charge density ($2 / \text{radius in nm}$) for each cation.

[2]

Cation	Ionic Radius / nm	Charge Density / nm ⁻¹	T(decomp) / K
Mg ²⁺	0.065	30.8	623
Ca ²⁺	0.099	20.2	834
Sr ²⁺	0.113	17.7	871
Ba ²⁺	0.135	14.8	923

- (b) Plot a graph of T_{decomp} against ionic radius and describe the relationship.

[2]

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- (c) Explain this trend in thermal stability in terms of cation polarising power.

[3]

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- (d) State the hazard associated with the gaseous product NO_2 and the safety precaution required.

[2]

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- (e) Suggest why anhydrous nitrates must be used rather than hydrated nitrates.

[1]

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QUESTION 8

PLANNING

[Total: 10 Marks]

Planning a Potentiometric Redox Titration of Iron(II) with Permanganate

Cambridge Syllabus Reference: 9701/51/M/J/20/Q1

Potentiometric titrations are used when solutions are cloudy or coloured, preventing visual indicator observation. Plan an experiment to determine the concentration of Fe^{2+} in an opaque mineral leachate using potentiometric titration with standardized $0.0200 \text{ mol dm}^{-3} \text{ KMnO}_4$ in acidic medium.

- (a) Draw a labelled diagram of the electrochemical apparatus including electrodes and voltmeter. [3]
- (b) State the role of the platinum electrode in this potentiometric measurement. [2]
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- (c) Describe how the titration should be carried out as the equivalence point is approached. [2]
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- (d) Explain how the exact equivalence volume is determined graphically from the titration data. [2]
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- (e) Explain why hydrochloric acid cannot be used to acidify the iron solution in this titration. [1]
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QUESTION 9

ANALYSIS & EVALUATION

[Total: 10 Marks]

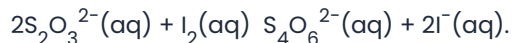
Analysis of Systematic Errors in the Iodometric Determination of Copper in Brass

Cambridge Syllabus Reference: 9701/52/O/N/24/Q2

Brass alloy was dissolved in concentrated nitric acid, neutralised with sodium carbonate, and acidified with ethanoic acid.

Excess KI was added to liberate iodine: $2\text{Cu}^{2+}(\text{aq}) + 4\text{I}^{-}(\text{aq}) \rightarrow 2\text{CuI}(\text{s}) + \text{I}_2(\text{aq})$.

The liberated iodine was titrated with $0.100 \text{ mol dm}^{-3} \text{Na}_2\text{S}_2\text{O}_3$ using starch indicator:



- (a) State at what point in the titration starch indicator must be added and explain why. [2]
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-
- (b) Copper(I) iodide, CuI, precipitates as a white precipitate that adsorbs iodine on its surface. State how potassium thiocyanate (KSCN) resolves this error. [2]
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-
- (c) Explain why nitric acid used to dissolve the brass must be completely removed or neutralized before adding KI. [2]
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-
- (d) Calculate the percentage apparatus uncertainty if burette reading is $24.60 \text{ cm}^3 (\pm 0.05 \text{ cm}^3)$ and mass of brass is $1.450 \text{ g} (\pm 0.001 \text{ g})$. [2]
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-
- (e) State the observation at the final stoichiometric endpoint in the presence of KSCN. [2]
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QUESTION 10

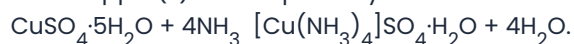
PLANNING

[Total: 10 Marks]

Planning the Synthesis and Anti-Solvent Precipitation of $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4 \cdot \text{H}_2\text{O}$

Cambridge Syllabus Reference: 9701/53/M/J/24/Q1

Tetraamminecopper(II) sulfate monohydrate, $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4 \cdot \text{H}_2\text{O}$, is a deep royal blue crystalline solid prepared from copper(II) sulfate pentahydrate and concentrated ammonia.



The complex is soluble in water but insoluble in ethanol.

- (a) Describe the colour and appearance changes observed as concentrated ammonia is added dropwise until in excess. [2]

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- (b) Explain the role of ethanol addition (anti-solvent precipitation) in this preparation. [2]

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- (c) Describe the filtration and washing technique to maximize yield and purity. [3]

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- (d) Calculate the percentage yield if 5.00 g of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ ($M_r = 249.6$) produces 4.12 g of $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4 \cdot \text{H}_2\text{O}$ ($M_r = 245.6$). [3]

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Comprehensive Examiner Mark Scheme

Marking points, indicative responses, and error-carried-forward (ECF) guidelines

MARK SCHEME: QUESTION 1 (PLANNING) — Planning a Colorimetric Determination of Complex Formula via Job's Method

[10 Marks]

Part	Examiner Marking Points / Indicative Content	Marks
(a)	Table showing total volume = 25.0 cm ³ for all runs [1]; equimolar solutions ensuring volume ratio equals mole ratio [1]; acidified with HNO ₃ to prevent Fe ³⁺ hydrolysis to Fe(OH) ₃ [1].	[3]
(b)	Use a blue / green-blue filter (approx 480–500 nm, complementary to blood-red colour) [1]; zero colorimeter using cuvette filled with distilled water / 0.1 mol dm ⁻³ HNO ₃ blank [1].	[2]
(c)	Plot absorbance on y-axis against x(SCN) on x-axis [1]; draw two intersecting lines of best fit through rising and falling points [1]; read x _{max} at intersection; $n = x_{\text{max}} / (1 - x_{\text{max}})$ [1].	[3]
(d)	$n = 0.50 / (1 - 0.50) = 1.0$ $n = 1$ [1]; formula: $[\text{Fe}(\text{SCN})]^{2+}$ (or $[\text{Fe}(\text{H}_2\text{O})_5(\text{SCN})]^{2+}$) [1].	[2]

MARK SCHEME: QUESTION 2 (ANALYSIS & EVALUATION) — Analysis and Evaluation of the Stoichiometry of Malachite Thermal Decomposition

[10 Marks]

Part	Examiner Marking Points / Indicative Content	Marks
(a)	Total mass of gases evolved = 44.0 + 18.0 = 62.0 g mol ⁻¹ [1]; % mass loss = $(62.0 / 221.0) \times 100\% = 28.05\%$ [1].	[2]
(b)	Mass lost = 4.420 - 3.280 = 1.140 g [1]; % mass loss = $(1.140 / 4.420) \times 100\% = 25.79\%$ [1].	[2]
(c)	1. Incomplete decomposition (crucible not heated strongly or long enough) [1]; 2. Residue reabsorbed moisture from air while cooling without desiccator [1].	[2]
(d)	Slightly ajar during heating to allow evolved CO ₂ and H ₂ O steam to escape while preventing spitting [1]; fully closed during cooling in desiccator to prevent reabsorption of atmospheric moisture [1].	[2]
(e)	Add dilute sulfuric acid: CuO dissolves forming blue solution without effervescence [1]; if unreacted carbonate is present, effervescence / gas turning limewater cloudy occurs [1].	[2]

MARK SCHEME: QUESTION 3 (PLANNING) — Planning a Redox Titrimetric Analysis of Chromium Content in Ferrochrome Alloy

[10 Marks]

Part	Examiner Marking Points / Indicative Content	Marks
(a)	To decompose and destroy all excess unreacted peroxodisulfate ions [1]; otherwise, excess S ₂ O ₈ ²⁻ would react with the standard Fe ²⁺ titrant, leading to an overestimation of chromium [1].	[2]
(b)	$\text{Cr}_2\text{O}_7^{2-} + 6\text{Fe}^{2+} + 14\text{H}^+ \rightarrow 2\text{Cr}^{3+} + 6\text{Fe}^{3+} + 7\text{H}_2\text{O}$ [2].	[2]
(c)	Barium diphenylamine sulfonate (or sodium diphenylamine sulfonate) [1]; colour change: violet / purple to green (or green to purple if titrating into Fe ²⁺) [1].	[2]
(d)	1. Moles Fe ²⁺ = conc × titre; 2. Moles Cr ₂ O ₇ ²⁻ = $(1/6) \times \text{moles Fe}^{2+}$ [1]; 3. Moles Cr in sample = 2 × moles Cr ₂ O ₇ ²⁻ [1]; 4. Mass Cr = moles Cr × 52.0 g mol ⁻¹ ; % Cr = $(\text{mass Cr} / \text{mass alloy}) \times 100\%$ [1].	[3]
(e)	Carcinogenic / mutagenic / toxic; wear nitrile gloves and handle in designated fume area [1].	[1]

MARK SCHEME: QUESTION 4 (PLANNING) — Planning the Synthesis and Recrystallisation of Hexaamminecobalt(III) Chloride

[10 Marks]

Part	Examiner Marking Points / Indicative Content	Marks
(a)	Acts as a heterogeneous catalyst for ligand exchange / ligand substitution [1]; facilitates replacement of aqua ligands by ammonia to favour formation of $[\text{Co}(\text{NH}_3)_6]^{3+}$ over pentaammine aquo complexes [1].	[2]
(b)	Cool mixture in ice bath to maximize precipitation of product [1]; filter crude solid mixture using Buchner funnel under reduced pressure [1].	[2]
(c)	Dissolve crude product in minimum volume of hot boiling dilute HCl (complex dissolves, charcoal remains insoluble) [1]; perform hot gravity filtration to remove insoluble charcoal [1]; cool filtrate in ice bath, precipitating pure orange crystals of $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$, and filter under vacuum [1].	[3]
(d)	Common ion effect (Cl ⁻) reduces solubility of $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ upon cooling, enhancing yield [1]; acidic medium prevents base hydrolysis of ammine ligands [1].	[2]
(e)	Wash with cold ethanol, then dry in a desiccator over anhydrous calcium chloride / vacuum oven at ≤ 50 °C [1].	[1]

MARK SCHEME: QUESTION 5 (ANALYSIS & EVALUATION) — Analysis and Evaluation of the Stability Constant (Kstab) of Copper(II) Ammine Complex**[10 Marks]**

Part	Examiner Marking Points / Indicative Content	Marks
(a)	Water is the solvent and its concentration is effectively constant ($\sim 55.5 \text{ mol dm}^{-3}$) [1]; it is incorporated into the equilibrium constant value [1].	[2]
(b)	$K_{\text{stab}} = (0.0480) / [(2.40 \times 10^{-14}) \times (0.250)^4] = 0.0480 / [9.375 \times 10^{-17}] = 5.12 \times 10^{14}$ [2]; units: $\text{mol}^{-4} \text{ dm}^{12}$ [1].	[3]
(c)	$G^\circ = -8.314 \times 298 \times \ln(5.12 \times 10^{14}) = -2477.6 \times 33.87 = -83.9 \text{ kJ mol}^{-1}$ [2].	[2]
(d)	1,2-diaminoethane is a bidentate ligand [1]; substitution of unidentate ligands by bidentate ligands increases total number of free particles in solution [1]; results in a large positive entropy change of reaction ($S_{\text{sys}} > 0$), driving G° significantly more negative [1].	[3]

MARK SCHEME: QUESTION 6 (PLANNING) — Planning Gravimetric Determination of Sulfate Content in Agricultural Fertilizer**[10 Marks]**

Part	Examiner Marking Points / Indicative Content	Marks
(a)	To prevent the co-precipitation of other insoluble barium salts such as BaCO_3 or $\text{Ba}_3(\text{PO}_4)_2$ which are acid-soluble [2].	[2]
(b)	Maintain the precipitate in contact with the hot mother liquor near $80\text{--}90^\circ\text{C}$ for 1 hour without boiling [1]; allows Ostwald ripening (dissolution of small particles and redeposition onto larger crystals), facilitating retention on filter paper and preventing colloidal loss [1].	[2]
(c)	Ashless filter paper (e.g. Whatman 42) [1]; use rubber-tipped glass rod ('policeman') and hot distilled water wash bottle to rinse all precipitate from beaker walls onto filter [1].	[2]
(d)	Heat crucible gently to char and burn off filter paper without flame, then heat strongly with blue Bunsen flame [1]; cool in desiccator, weigh, and repeat heating until two successive weighings agree within $\pm 0.002 \text{ g}$ [1].	[2]
(e)	Add a few drops of BaCl_2 solution to the clear supernatant / filtrate [1]; absence of cloudiness confirms complete precipitation [1].	[2]

MARK SCHEME: QUESTION 7 (ANALYSIS & EVALUATION) — Quantitative Analysis of Decomposition Trends in Group 2 Nitrates**[10 Marks]**

Part	Examiner Marking Points / Indicative Content	Marks
(a)	Charge density values correct: 30.8, 20.2, 17.7, 14.8 nm^{-1} [2].	[2]
(b)	Straight/smooth curve plotted with positive slope [1]; as ionic radius increases, decomposition temperature increases (thermal stability increases down Group 2) [1].	[2]
(c)	Smaller cations have higher charge density / higher polarising power [1]; Mg^{2+} distorts the electron cloud of the nitrate anion (N-O bond) more strongly than Ba^{2+} [1]; weakens the N-O bond, requiring less thermal energy to break the bond and decompose [1].	[3]
(d)	NO_2 is toxic / severe respiratory irritant [1]; carry out heating inside a well-ventilated fume cupboard [1].	[2]
(e)	Hydrated nitrates release steam which condenses and creates thermal shock cracking hot boiling tube / causes basic nitrate hydrolysis [1].	[1]

MARK SCHEME: QUESTION 8 (PLANNING) — Planning a Potentiometric Redox Titration of Iron(II) with Permanganate**[10 Marks]**

Part	Examiner Marking Points / Indicative Content	Marks
(a)	Burette containing standard KMnO_4 clamped over titration beaker [1]; platinum indicator electrode and reference electrode (calomel / Ag-AgCl) immersed in solution [1]; high-resistance digital voltmeter / pH-meter connected across electrodes [1].	[3]
(b)	Inert electron conductor that does not participate chemically in the reaction [1]; adopts potential determined by ratio of oxidised and reduced forms in solution: $[\text{Fe}^{3+}]/[\text{Fe}^{2+}]$ or $[\text{MnO}_4^-]/[\text{Mn}^{2+}]$ [1].	[2]
(c)	Add KMnO_4 in 1.0 cm^3 increments initially; when rapid potential changes occur near equivalence, add titrant dropwise (0.05 or 0.10 cm^3 increments) while stirring constantly [2].	[2]
(d)	Plot cell potential E against volume V (inflection point of S-shaped curve) OR plot first derivative E/V against V where peak maximum corresponds to exact equivalence volume [2].	[2]
(e)	KMnO_4 is a strong enough oxidising agent to oxidise Cl^- to toxic chlorine gas, causing an erroneously high titre [1].	[1]

MARK SCHEME: QUESTION 9 (ANALYSIS & EVALUATION) — Analysis of Systematic Errors in the Iodometric Determination of Copper in Brass**[10 Marks]**

Part	Examiner Marking Points / Indicative Content	Marks
(a)	Add starch when solution becomes pale straw-yellow (near endpoint) [1]; if added too early, iodine binds irreversibly to starch complex, preventing sharp decolorisation and underestimating titre [1].	[2]
(b)	Add KSCN near endpoint; SCN ⁻ replaces adsorbed I ₂ on CuI surface (forming CuSCN) [1]; releases bound iodine into solution, ensuring complete reaction with thiosulfate and a sharp milky-white endpoint [1].	[2]
(c)	Residual HNO ₃ / NO ₂ ⁻ oxidises iodide ions to iodine (2I ⁻ + 4H ⁺ + 2NO ₂ ⁻ → I ₂ + 2NO + 2H ₂ O) [1]; causes extra iodine liberation, resulting in falsely high copper percentage [1].	[2]
(d)	Burette % uncertainty = $(0.10 / 24.60) \times 100\% = 0.407\%$ [1]; Balance % uncertainty = $(0.002 / 1.450) \times 100\% = 0.138\%$; Total = 0.55% [1].	[2]
(e)	Disappearance of blue-black colour leaving a persistent milky-white / cream precipitate of CuSCN [2].	[2]

MARK SCHEME: QUESTION 10 (PLANNING) — Planning the Synthesis and Anti-Solvent Precipitation of [Cu(NH₃)₄]SO₄·H₂O**[10 Marks]**

Part	Examiner Marking Points / Indicative Content	Marks
(a)	Pale blue solution forms light blue precipitate of Cu(OH) ₂ [1]; with excess ammonia, precipitate dissolves to give a deep royal blue solution [1].	[2]
(b)	Ethanol decreases the dielectric constant of the solvent mixture [1]; lowers the solubility of the ionic complex, causing [Cu(NH ₃) ₄]SO ₄ ·H ₂ O crystals to precipitate rapidly [1].	[2]
(c)	Filter using Buchner funnel and suction flask under vacuum [1]; wash crystals with equal-volume mixture of ethanol and concentrated ammonia (to prevent ligand loss) [1]; final wash with cold diethyl ether to facilitate drying [1].	[3]
(d)	Moles CuSO ₄ ·5H ₂ O = $5.00 / 249.6 = 0.02003$ mol [1]; Theoretical mass = $0.02003 \times 245.6 = 4.920$ g [1]; % yield = $(4.12 / 4.920) \times 100\% = 83.7\%$ [1].	[3]